Introduction to Compressed Sensing. Algorithmic complexity and convex relaxation

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Abstract

In these lecture notes we introduce the ℓ_0 minimization procedure, and we show that it is optimal from a theoretical point of view. We also show that it cannot be used in practice, because it is an NP-hard procedure in general.

We then introduce the Basis Pursuit procedure by convex relaxation. We show that it can be written as a linear programming problem. Finally, we show that if Basis Pursuit solves the CS problem for the set of all s-sparse vectors, then the number of measurements must necessarily be larger than $s\log(eN/s)$.

1 Introduction

The CS problem is stated as follows: recover a vector $x \in \mathbb{R}^N$ from the measurements y $(= Ax)$ and from the matrix of measurement vectors $A \in \mathbb{R}^{m\times N}$, knowing that x has a support of small size.

We therefore wish to solve a system of the form given in Figure 1, in which there are far more unknowns (here N) than equations (here m).

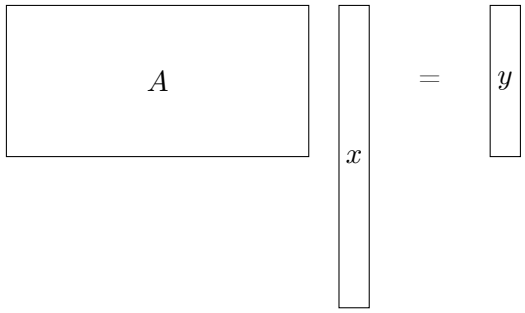


Figure 1: Recover x from $y = Ax$ and $A \in \mathbb{R}^{m\times N}$, where $m < N$, knowing that x is sparse.

The central assumption in CS is that the signal x to be reconstructed is **sparse**. A natural idea is therefore to look, in the affine space of solutions of the linear system $Ax = y$,

$$\{t \in \mathbb{R}^N : At = y\} = x + \mathrm{Ker}(A)$$

for the sparsest element, that is, the one whose support has the smallest size. This procedure is called the ℓ_0 minimization procedure, which we now introduce.

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Definition 1.1 (ℓ_0 minimization procedure)**.** *The ℓ_0* ***minimization procedure*** *is the map which, to every $y \in \mathbb{R}^m$ and every $A \in \mathbb{R}^{m \times N}$, returns*

$$\hat{x}_0 \in \operatorname*{argmin}_{t \in \mathbb{R}^N : At = y} \|t\|_0 \tag{P_0}$$

if the equation $Ax = y$ admits a solution, and $\emptyset$ otherwise. We recall that $\|t\|_0 = |\mathrm{supp}(t)|$ is the size of the support of t.

The linear system presented in Figure 1 admits the whole affine space $x + \mathrm{Ker}(A)$ as its set of solutions. Since $m < N$, A has a kernel of dimension at least $N - m \geq 1$. There are therefore infinitely many solutions to this system of equations.

On the other hand, we have an additional piece of information: the vector x that we seek to recover has a support of small size. The ℓ_0 minimization procedure is therefore built so as to search, in the space of solutions of the system in Figure 1, for the solution with the smallest support.

The hope is that the kernel of A will be "well positioned", so that the solution space $x + \mathrm{Ker}(A)$ intersects the set of s-sparse vectors Σ_s, where $s = \|x\|_0$, only at x:

$$\big(x + \mathrm{Ker}(A)\big) \cap \Sigma_s = \{x\} \text{ for } s = \|x\|_0$$

where we write

$$\Sigma_s = \left\{ x \in \mathbb{R}^N : \|x\|_0 \leq s \right\}.$$

The following proposition shows that this condition is necessary and sufficient for the ℓ_0 minimization procedure to reconstruct the vector x exactly.

Proposition 1.2. *Let $x \in \mathbb{R}^N$. Write $s = \|x\|_0$. The following two assertions are equivalent:*

1. *x is the unique solution of the ℓ_0 minimization problem, that is, $\hat{x}_0 = x$*
2. *$\big(x + \mathrm{Ker}(A)\big) \cap \Sigma_s = \{x\}$.*

Proof. If x is the unique solution of the ℓ_0 minimization problem, then for every $t \neq x$ in the solution space $\big(x + \mathrm{Ker}(A)\big)$ one has $\|t\|_0 > s$, and therefore $\big(x + \mathrm{Ker}(A)\big)$ intersects Σ_s only at x.

Conversely, if $\big(x + \mathrm{Ker}(A)\big)$ intersects Σ_s only at x, then every vector $t \neq x$ in $\big(x + \mathrm{Ker}(A)\big)$ satisfies $\|t\|_0 > s$. The vector with the smallest support in $\big(x + \mathrm{Ker}(A)\big)$ is therefore x, and it is the only vector with this property. Hence the ℓ_0 minimization algorithm admits a unique solution, which is x, i.e. $\hat{x}_0 = x$. ∎

Proposition 1.2 is the first result we encounter in which the role of the "orientation" of the kernel $\mathrm{Ker}(A)$ with respect to the set of s-sparse vectors is brought to the fore.

The idea behind Proposition 1.2 — which will be present in all the reconstruction results of this course — is that the space $\mathrm{Ker}(A)$ must be "far" from the set Σ_s. Geometrically, one wants $\mathrm{Ker}(A)$ to intersect Σ_s only at 0, and to lie "close to diagonal elements of $\mathbb{R}^N$" (i.e., the "least sparse" elements of $\mathbb{R}^N$).

Proposition 1.2 is not, however, a typical result in CS. Indeed, Proposition 1.2 gives a necessary and sufficient condition for the exact reconstruction, by a procedure (here ℓ_0 minimization), of a single vector x. In the remainder of the course, we shall be interested only in properties of the measurement matrix that make it possible to reconstruct every vector in Σ_s.

2 Minimal number of measurements

We are interested here in the following question: given $A \in \mathbb{R}^{m\times N}$, the system in Figure 1 is highly underdetermined, since $m < N$ (and even $m << N$); nevertheless, knowing that one seeks a solution x that is s-sparse, what is the minimal number of measurements m (i.e., of rows of A) required in order to reconstruct every vector in Σ_s from only m measurements?

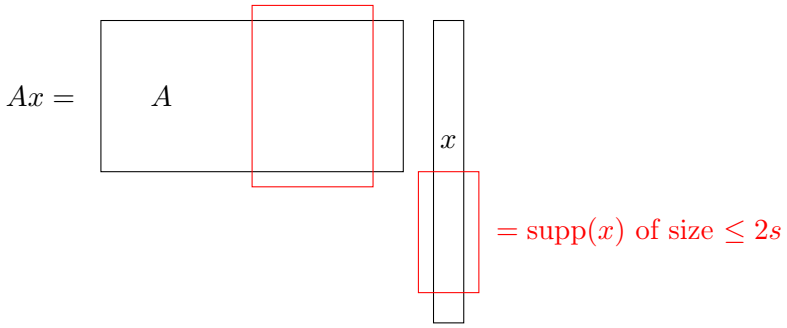
We stress that we wish to be able to reconstruct every vector in Σ_s (and not merely a single vector, as in Proposition 1.2). In particular, it is necessary that $Ax \neq Ay$ for all $x, y \in \Sigma_s$ if one is to have any chance of recovering x (or y) from Ax (or Ay). This necessary condition has a cost in terms of the number of measurements and of the "orientation" of $\mathrm{Ker}(A)$.

Proposition 2.1. *The following assertions are equivalent:*

1. *for all $x, y \in \Sigma_s$, if $x \neq y$ then $Ax \neq Ay$*
2. $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$ *(and hence $m \geq 2s$)*
3. *every submatrix of A of size $m \times (2s)$ is injective (and hence $m \geq 2s$).*

Proof. If, for all $x, y \in \Sigma_s$, one has $Ax \neq Ay$ whenever $x \neq y$, then $A(x-y) \neq 0$, and therefore the only $2s$-sparse vector in $\mathrm{Ker}(A)$ is the zero vector (because every $2s$-sparse vector can be written as the difference of two s-sparse vectors). In other words, $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$. This condition does imply that $m \geq 2s$: if one extracts from A the matrix of size $m \times (2s)$ made up of the first $2s$ columns of A, one sees that it is injective and, in particular, that its number of rows must be at least its number of columns, i.e. $m \geq 2s$.

We now assume that $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$. Let $J \subset \{1, \ldots, N\}$ be such that $|J| = 2s$. Let $x \in \mathbb{R}^N$ be nonzero and supported on J. In particular, x is $2s$-sparse. Then, by assumption, $Ax \neq 0$, i.e. $A_{\cdot J} x_J \neq 0$ (cf. Figure 2), and therefore, for every nonzero $y \in \mathbb{R}^J$, $A_{\cdot J} y \neq 0$. In other words, $A_{\cdot J}$ is injective. Since this holds for every $J \subset \{1, \ldots, N\}$ with $|J| = 2s$, we conclude that every extracted submatrix of A of size $m \times (2s)$ is injective.



Finally, we assume that every matrix of size $m \times (2s)$ extracted from A is injective. Let $x, y \in \Sigma_s$ be such that $x \neq y$. Write $J_x = \mathrm{supp}(x)$ and $J_y = \mathrm{supp}(y)$; then $|J_x \cup J_y| \leq 2s$. In particular, $A_{\cdot J_x \cup J_y}$ is injective, and since $x - y \neq 0$ and $\mathrm{supp}(x-y) \subset J_x \cup J_y$, one has $A(x-y) = A_{\cdot J_x \cup J_y}(x-y) \neq 0$. Hence $Ax \neq Ay$. ∎

One conclusion of Proposition 2.1 is that if one wants any chance of reconstructing exactly the vectors in Σ_s, then it is necessary to have at least $2s$ measurements. We now show that this condition is sufficient for the ℓ_0 minimization algorithm when the kernel of the matrix A is well positioned.

Proposition 2.2. *Let $A \in \mathbb{R}^{m\times N}$. The following two assertions are equivalent:*

1. $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$

2. *the ℓ_0 minimization procedure reconstructs exactly every vector in Σ_s:*

$$\underset{t \in \mathbb{R}^N : At = Ax}{\operatorname{argmin}} \|t\|_0 = \{x\} \text{ for all } x \in \Sigma_s.$$

Proof. Let $x \in \Sigma_s$. Let $t \in \mathbb{R}^N$ be such that $At = Ax$. One has $A(x-t) = 0$. Since $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$, either $x = t$ or $|\mathrm{supp}(x-t)| > 2s$. As $|\mathrm{supp}(x)| \leq s$, the condition $|\mathrm{supp}(x-t)| > 2s$ necessarily implies that $|\mathrm{supp}(t)| \geq s+1$, i.e. $\|t\|_0 > \|x\|_0$. Thus either $x = t$ or $\|t\|_0 > \|x\|_0$, which implies that the vector with the smallest support in the solution space $x + \mathrm{Ker}(A)$ is x.

For the converse, let $t \in \mathrm{Ker}(A) \cap \Sigma_{2s}$. We want to show that $t = 0$. We write $t = x - z$ where $\|x\|_0 = \|z\|_0 \leq s$. Since $x \in \Sigma_s$, x is the unique solution to (P): $\min_{t \in \mathbb{R}^N : At = Ax} \|t\|_0$. Hence, there is no solution to (P) that is sparser than $\|x\|_0$. But, $\|x\|_0 = \|z\|_0$ and $Az = Ax$, hence, z is also a solution to (P) and so by uniqueness of this solution $z = x$ and so $t = 0$. ∎

We thus see that the condition $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$ is central both to the feasibility of the problem (cf. Proposition 2.1) and to its realization by the ℓ_0 minimization algorithm (cf. Proposition 2.2). The question that naturally comes to mind concerns the existence of such matrices: is it possible to construct matrices $A \in \mathbb{R}^{m \times N}$ such that $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$? Moreover, we wish to take as few measurements as possible. We would therefore like to construct such matrices for a minimal number of measurements m. We already know that this property cannot hold for $m < 2s$ (cf. Proposition 2.1). The following result shows that this number is in fact sufficient.

Proposition 2.3. *For every $N \geq 2s$, there exists a matrix $A \in \mathbb{R}^{m \times N}$ such that:*

1. $m = 2s$

2. $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$.

Proof. We use Vandermonde matrices for this construction. Let $t_N > \cdots > t_1 > 0$. Set

$$A = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ t_1 & t_2 & \cdots & t_N \\ \vdots & \vdots & \vdots & \vdots \\ t_1^{2s-1} & t_2^{2s-1} & \cdots & t_N^{2s-1} \end{pmatrix}.$$

Let $J \subset \{1, \ldots, N\}$ be of size $|J| = 2s$. Write $J = \{j_1, \ldots, j_{2s}\}$, where $j_{2s} > \cdots > j_1$. To show that the extracted matrix $A_{\cdot J}$ is invertible, it suffices to compute its determinant. One has

$$\det(A_{\cdot J}) = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ t_{j_1} & t_{j_2} & \cdots & t_{j_N} \\ \vdots & \vdots & \vdots & \vdots \\ t_{j_1}^{2s-1} & t_{j_2}^{2s-1} & \cdots & t_{j_N}^{2s-1} \end{vmatrix} = \prod_{k<l} (t_{j_k} - t_{j_l}) > 0.$$

The matrix $A_{\cdot J}$ is therefore invertible. Since this holds for every $|J| = 2s$, we deduce from Proposition 2.1 that $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$. ∎

Proposition 2.3 tells us how to choose the measurements that must be taken in order to solve the CS problem with a minimal number of observations. If A is a Vandermonde matrix (cf. the proof of Proposition 2.3) of size $(2s) \times N$, then the ℓ_0 minimization algorithm reconstructs every vector in Σ_s from $2s$ measurements when the measurement matrix is A.

Conclusion: $m = 2s$ is the number of measurements that is necessary and sufficient in order to reconstruct exactly every vector in Σ_s from m linear measurements. In this case, the ℓ_0 minimization procedure provides an exact reconstruction procedure whenever the measurement matrix $A \in \mathbb{R}^{m \times N}$ satisfies $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$. Such matrices exist for $m = 2s$.

The approach followed in this section is typical in CS:

1. one fixes a procedure (here, the ℓ_0 minimization method)
2. one identifies one or several properties of the measurement matrix $A \in \mathbb{R}^{m \times N}$ which ensure that the procedure yields the exact reconstruction of the vectors in Σ_s
3. one constructs matrices A satisfying this property, or these properties, and requiring only a minimal number of rows — that is, a minimal number of measurements m.

In the sequel we shall study other procedures (in particular, Basis Pursuit), for which we shall identify conditions on A that allow these procedures to solve the CS problem on Σ_s (in particular, the RIP condition). We shall then construct matrices satisfying these conditions for a minimal number of measurements (random matrices will satisfy the RIP for $m \sim s \log(N/s)$).

In view of the results of this section and of the conclusion drawn from them, the CS problem (recovering every vector $x \in \Sigma_s$ from Ax and A) appears to be solved by the ℓ_0 minimization method, for a minimal number of measurements $m = 2s$ and, for instance, by means of measurement vectors built from Vandermonde matrices of size $(2s) \times N$.

This is not the case, however, because the ℓ_0 minimization procedure cannot, in general, be implemented in practice. We have therefore proposed a procedure which is of theoretical interest only, and which is not used in practice. This is what we study in the next section.

3 The NP-hard property of ℓ_0 minimization

In this section we consider one of the most important aspects of CS and, more generally, of modern statistics: the "computational" aspect of procedures. We shall therefore be concerned with the implementation and the feasibility of procedures.

So far, we have encountered only the ℓ_0 minimization procedure. As seen in the previous section, this procedure solves the CS problem in an optimal way, but can it be used in practice?

Remark 3.1 (Procedures and algorithms)**.** *Throughout the course, we distinguish between procedures and algorithms. A procedure is, broadly speaking, a statistic, that is, a (measurable) function of the observations. This notion contains no computational aspect and is not concerned with practical use. An algorithm is associated with a procedure: it is a computational routine (usually described by pseudocode) that gives a concrete way of implementing (i.e., of coding) the procedure.*

There are two kinds of algorithms: those that can be used in practice, that is, which do not require too much computation time (or too large a computational capacity), and those that cannot be used in practice because of limitations in computational power or computation time.

The ideal, for a statistician, is to propose a procedure which enjoys optimality properties from a statistical and theoretical point of view (for instance, being minimax) and for which there exists an algorithm that can be used in practice.

One algorithmic way of solving the ℓ_0 minimization problem is to proceed as follows.

Data: $y \in \mathbb{R}^m, A \in \mathbb{R}^{m \times N}$
Output: solution $\hat{x}_0$ of (P_0)
for $s = 1, 2, 3, \ldots,$ **do**
  for $J \subset \{1, \ldots, N\}$ *of size* $|J| = s$ **do**
    Solve the linear system $A_{\cdot J} x = y$ of size $m \times s$.
    if *There is a solution* **then**
      record this solution
      exit the loops
    else
      continue
    end
  end
end

Algorithm 1: Algorithm for solving the ℓ_0 minimization procedure.

Under the assumption that the ℓ_0 minimization procedure "works", that is, when $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$, the first set J for which the system $A_{\cdot J} x = y$ admits a solution is $J = \mathrm{supp}(x)$, where $x \in \Sigma_s$ is the sparsest solution in $x + \mathrm{Ker}(A)$. This is therefore the unique solution of the ℓ_0 minimization problem, and it is the solution returned by Algorithm 1.

This algorithm thus implements the ℓ_0 minimization procedure in the regime where this procedure applies, that is, under the assumption that $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$.

In a few words, Algorithm 1 proceeds as follows. Given a number s of columns, it carries out a search over all sets $J \subset \{1, \ldots, N\}$ of size s, looking for the first set J such that $A_{\cdot J} x = y$ admits a solution. For s fixed, if such a J exists, then Algorithm 1 solves the system $A_{\cdot J} x = y$ and returns the solution. If there is no solution on s columns, then the algorithm iterates on s (that is, $s \to s + 1$), and this continues until a support J is found such that $A_{\cdot J} x = y$ (such a support exists, since $y = Ax^*$ for some x^*).

The weak point of this algorithm is the step in which one searches over all sets J of size s in $\{1, \ldots, N\}$. There are

$$\begin{pmatrix} N \\ s \end{pmatrix}$$

possible choices. For instance, for $N = 1000$ and $s = 10$, one has

$$\binom{N}{s} = \frac{N(N-1)\cdots(N-s+1)}{s(s-1)\cdots 1} \geq \left(\frac{N}{s}\right)^s = 10^{20}$$

possible choices. For each set $J \subset \{1, \ldots, N\}$, Algorithm 1 solves the linear system $A_{\cdot J} x = y$. If solving one such system takes 10^{-10} seconds, then the computation time for the step $s = 10$ of the algorithm is 10^{10} seconds, that is, 300 years.

This algorithm therefore cannot be executed in a reasonable amount of time. It is consequently not used in practice. Its weak point is this search over all subsets of a given cardinality inside a larger set. This type of exploration is a combinatorial search, which lies at the heart of many algorithms that run in exponential time, that is, algorithms for which an exponential number of steps, in the dimension of the inputs, is required in order for them to be executed.

This is the case of Algorithm 1: its execution requires a number of steps of order

$$\sum_{j=1}^{s} \binom{N}{j} \sim \left(\frac{N}{s}\right)^s$$

which is exponential in s. Even for reasonable values of s and N (such as $s \sim 10$ and $N \sim 1000$), this number is already too large.

One may now ask about the efficiency of Algorithm 1. Perhaps there exists a more intelligent way of implementing the ℓ_0 minimization procedure than the one used by Algorithm 1 (which relies on a combinatorial search)?

The main result of this section is that the ℓ_0 minimization method is fundamentally difficult to implement (and hence unusable in practice). Algorithm 1 is therefore not so naive in general (cf. Remark 3.3 for a particular case in which the ℓ_0 minimization method can be implemented more rapidly). To this end, we must introduce, in an informal way, a few notions of algorithmic complexity.

There are several classes of decision problems:

1. $\mathfrak{P}$: *the set of all problems for which there exists an algorithm that can be executed in polynomial time.* For instance, the algorithm which, to a pair (A, B) of $n \times n$ matrices, associates the product AB, can be carried out in time $n^2 \times (n+n-1) = \mathcal{O}(n^3)$ (one fills the n^2 entries of AB by performing n multiplications and $n-1$ additions). For background, there exists an algorithm running in time $\mathcal{O}(n^{2.3728596})$ that solves this problem, but it is not known (as of 2021) whether one can do better.

2. $\mathfrak{NP}$: *the set of all problems for which there exists a polynomial-time algorithm certifying a solution.* For instance, if one considers the problem of the existence of a clique (that is, a fully connected subgraph) of size k inside a graph, then, given a subgraph on k vertices, one can easily check whether it is a clique in polynomial time (one simply checks that each node is connected to all the others; this therefore takes at most $\mathcal{O}(k^2)$).

3. $\mathfrak{NP}$-hard: *the set of all problems for which there exists an algorithm solving the problem that can be transformed, in polynomial time, into another algorithm solving any NP problem.* Intuitively, this is the set of all problems that are as hard to solve as any NP problem.

4. $\mathfrak{NP}$-complete: *the set of all problems that belong both to the class $\mathfrak{NP}$ and to the class $\mathfrak{NP}$-hard.* This is therefore the class of all NP problems that are as hard to solve as any other NP problem.

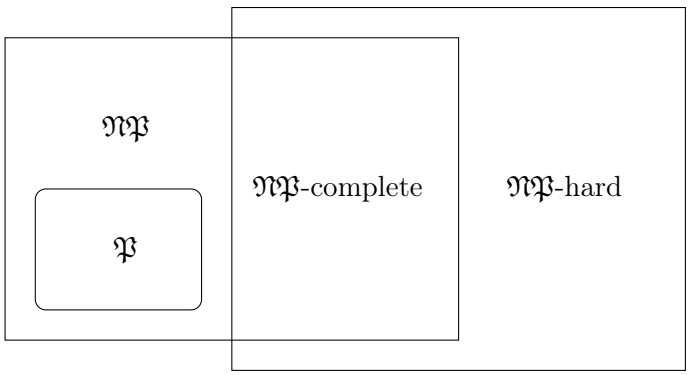


Figure 2: Classes of problems in decision theory

There exists a catalogue of NP-complete problems (there are currently on the order of 3000 referenced NP-complete problems, of which 6 are fundamental). These problems are particularly useful, because it "suffices" to find an algorithm solving one of them in order to solve all the others

in the class $\mathfrak{NP}$. They are, in a sense, regarded as the most complicated NP problems. In particular, if one succeeds in reducing (in polynomial time) a given problem to an NP-complete problem, then this problem is NP-hard. This is the strategy we shall use to show that ℓ_0 minimization is NP-hard.

We shall therefore reduce the ℓ_0 minimization problem to an NP-complete problem. To this end, we give a few examples of NP-complete problems, including the one that will be used in our reduction of problem (P_0).

Probably the best-known NP-complete problem is the **traveling salesman problem**: given a list of cities and the distances between them, find the shortest tour that passes through every city and returns to its starting point.

Another fairly well-known NP-complete problem is that of **finding a maximum clique**: given an undirected graph, find a clique (that is, a subgraph in which any two vertices are adjacent) of maximal size. Establishing properties of graphs is a good source of NP-complete problems. There are also partitioning problems: given integers $a_1, \dots, a_n$, the problem is to decide whether there exists a partition $I \sqcup J = \{1, \dots, n\}$ such that $\sum_{i \in I} a_i = \sum_{j \in J} a_j$. This decision problem is NP-complete.

Other well-known NP-complete problems are **graph coloring, the knapsack problem, and so on.**

The NP-complete problem of interest to us here is that of **partitioning by sets of cardinality** 3: Let $\{\mathcal{C}_j : j \in \{1, \dots, N\}\}$ be a collection of subsets of $\{1, \dots, m\}$, each of cardinality equal to 3. Does there exist a partition of $\{1, \dots, m\}$ made up only of elements of the collection $\{\mathcal{C}_j : j \in \{1, \dots, N\}\}$? (In mathematical terms: does there exist $I \subset [N]$ such that for all $i, j \in I$, if $i \neq j$ then $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset$, and $\cup_{i \in I} \mathcal{C}_i = \{1, \dots, m\}$?)

One easily checks that the problem of partitioning by sets of cardinality 3 lies in NP. Indeed, if one is given a candidate solution $\{\mathcal{C}_j : j \in J\}$, where $J \subset \{1, \dots, N\}$, it suffices to check first that $3|J| = m$ and that if $i, j \in J$ then $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset$. All of these operations can be carried out in time $\mathcal{O}(m^2)$. We take as given, however, that the problem of partitioning by sets of cardinality 3 is NP-complete (the proof is rather long, because it consists in showing that the problem *of partitioning by sets of cardinality* 3 "reduces" — that is, is equivalent up to rewriting — to the NP-complete problem "*3DM*", which itself reduces to the problem "*3-SAT*", which belongs to the "core 6" NP-complete problems of theoretical computer science). We shall reduce the ℓ_0 minimization problem to this problem. Here, "reduce" means transforming one problem into another by a procedure that can be carried out in polynomial time.

Theorem 3.2. *The ℓ_0 minimization procedure is NP-hard in general.*

Proof. We reduce the ℓ_0 minimization problem to the problem of partitioning by sets of cardinality 3.

Let $\{\mathcal{C}_j : j \in \{1, \dots, N\}\}$ be a collection of subsets of $\{1, \dots, m\}$ of cardinality 3. We first observe that one may assume that m is divisible by 3; otherwise, it is clear that $\{1, \dots, m\}$ cannot be partitioned by elements of $\{\mathcal{C}_j : j \in \{1, \dots, N\}\}$, since all of them have cardinality 3. We therefore assume in what follows that 3 divides m.

We define vectors $a_1, \dots, a_N$ in $\mathbb{R}^m$ by

$$(a_j)_i = \begin{cases} 1 & \text{if } i \in \mathcal{C}_j \\ 0 & \text{otherwise.} \end{cases}$$

Thus a_j is the indicator vector in $\mathbb{R}^m$ of $\mathcal{C}_j$: $a_j = I_{\mathcal{C}_j}$, and the columns of A are the indicators of the sets $\mathcal{C}_j$ for $j = 1, \dots, N$: $A = [a_1 | \cdots | a_N] \in \mathbb{R}^{m \times N}$, with columns the vectors a_j, $j = 1, \dots, N$. Finally, we construct a vector $y = (1, \dots, 1)^\top \in \mathbb{R}^m$.

The construction of A and y can be carried out in polynomial time, since $N \leq \binom{m}{3}$ (the cardinality of the collection $\{\mathcal{C}_j : j \in \{1, \dots, N\}\}$ is necessarily at most the number of all subsets of cardinality 3 in $\{1, \dots, m\}$).

We begin by observing that if $x \in \mathbb{R}^N$ is such that $Ax = y$, then, since $Ax = \sum_{j=1}^N x_j a_j$ and $\|a_j\|_0 = 3$ for every $j = 1, \dots, N$, the vector Ax necessarily has at most $3\|x\|_0$ nonzero coordinates. Therefore, since $\|Ax\|_0 = \|y\|_0 = m$, one has $\|x\|_0 \geq m/3$ (we recall that 3 divides m).

We now run the ℓ_0 minimization algorithm on the measurement matrix A and the measurement vector y. We obtain a vector $\hat{x}_0$ of minimal cardinality among all $x \in \mathbb{R}^N$ such that $Ax = y$ (i.e., $\hat{x}_0 \in \operatorname{argmin}_{Ax=y} \|x\|_0$). One has $A\hat{x}_0 = y$ and therefore, by what was established above, necessarily $\|\hat{x}_0\|_0 \geq m/3$. Two cases may then be considered:

1. if $\|\hat{x}_0\|_0 = m/3$, then the collection $\hat{\mathcal{C}}_0 := \{\mathcal{C}_j : j \in \operatorname{supp}(\hat{x}_0)\}$ forms a partition of $\{1, \dots, m\}$. Indeed, there are exactly $m/3$ sets (of cardinality 3) in $\hat{\mathcal{C}}_0$. Hence, if two sets in $\hat{\mathcal{C}}_0$ failed to be disjoint, then necessarily one of the coordinates of $A\hat{x}_0$ would vanish (because $A\hat{x}_0$ would be the sum of $m/3$ indicator vectors, each with a support of cardinality 3). This is not the case, however, since $A\hat{x}_0 = y$ has no zero coordinate. Therefore $\hat{\mathcal{C}}_0$ consists of $m/3$ disjoint sets of cardinality 3. It thus forms a partition of $\{1, \dots, m\}$.

2. if $\|\hat{x}_0\| > m/3$, then there is no partition of $\{1, \dots, m\}$ within the collection $\{\mathcal{C}_j : j \in \{1, \dots, N\}\}$. Indeed, if such a partition $\{\mathcal{C}_j : j \in J\}$ existed, then the vector $z \in \mathbb{R}^N$ defined by $z_j = 1$ if $j \in J$ and $z_j = 0$ if $j \notin J$ would satisfy $Az = y$ and $\|z\|_0 = m/3 < \|\hat{x}_0\|_0$, which contradicts the ℓ_0-minimality of $\hat{x}_0$.

One can therefore reduce, in polynomial time, an algorithm solving (P_0) to an algorithm solving the problem of partitioning by sets of cardinality 3. Since this problem is NP-complete, (P_0) is NP-hard. ■

Conclusion: The ℓ_0 minimization procedure is optimal from a theoretical point of view. It solves the CS problem for a minimal number of measurements ($m = 2s$) and under the minimal assumption on the measurement matrix ($\operatorname{Ker}(A) \cap \Sigma_{2s} = \{0\}$). Moreover, it is possible to construct measurement matrices satisfying this condition with a minimal number of rows, $m = 2s$. The ℓ_0 minimization procedure cannot, however, be used in practice in general, because it is an NP-hard procedure. It is therefore satisfactory only from a theoretical point of view, and not from a practical one.

Remark 3.3. *The phrase "in general", as used in Theorem 3.2, means that problem (P_0) is NP-hard in general. It may nevertheless happen that, in certain particular cases, this problem can be solved efficiently. The aim of this remark is to give one such example.*

In the particular case of Fourier measurements, it is possible to implement the ℓ_0 minimization algorithm very rapidly (in linear time). This is "Prony's algorithm".

We now introduce the setting in which "Prony's algorithm" is effective. We shall not present the algorithm itself, because understanding it requires some tools from Fourier analysis that lie outside the scope of this course.

Given a signal $x \in \mathbb{C}^N$ and a set $I \subset \{1, \dots, N\}$ of frequencies, it is in general impossible to reconstruct x from the Fourier coefficients $(\hat{x}_i : i \in I)$ alone, unless $I = \{1, \dots, N\}$ (because

the Fourier transform is a bijection of $\mathbb{C}^N$). But when one knows that x is sparse, the problem becomes feasible even for sets I of cardinality $2s$. Measurements of this so-called "structured" type are widely used in practice.

To fix the notation, we introduce the Fourier matrix (the discrete Fourier transform):

$$\Gamma : \left\{ \begin{array}{ccc} \mathbb{C}^N & \to & \mathbb{C}^N \\ x & \to & \Gamma x = \hat{x} \end{array} \right. \quad \text{where } \Gamma = \Big(\frac{w^{(p-1)(q-1)}}{\sqrt{N}} \Big)_{1 \leq p,q \leq N} \text{ and } w = \exp(-2\mathbf{i}\pi/N). \tag{3.1}$$

We denote by $\bar{\Gamma}_1, \ldots, \bar{\Gamma}_N$ the row vectors of Γ. One therefore has $\hat{x}_i = \langle \Gamma_i, x \rangle$ for every $i = 1, \ldots, N$.

For a subset $I \subset \{1, \ldots, N\}$ of frequencies, the data at hand are $(\hat{x}_i : i \in I)$. In matrix form, one is given $\Gamma_I x$, where

$$\Gamma_I : \left\{ \begin{array}{ccc} \mathbb{C}^N & \to & \mathbb{C}^{|I|} \\ x & \to & (\hat{x}_i : i \in I). \end{array} \right.$$

This matrix is thus the matrix of measurement vectors in Compressed Sensing.

"Prony's algorithm" recovers x from the first $2s$ Fourier coefficients of x, and this for every $x \in \Sigma_s$. The procedure therefore solves the CS problem optimally (for a number of measurements equal to $2s$) and can be implemented by a linear-time algorithm.

This is an example of an ideal situation in statistics: a procedure that is theoretically optimal and implementable in linear time. Prony's algorithm is, however, neither robust (with respect to noise) nor stable (when the signal is not exactly sparse, the algorithm fails). This is why it is used in practice only for certain very reliable data sets (such as DVDs).

4 Convex relaxation and the Basis Pursuit procedure

The ℓ_0 minimization procedure cannot be used in practice, because of its NP-hardness. One must therefore find another procedure for which a polynomial-time algorithm can be constructed, and for which one can prove theoretical properties close to those of (P_0).

For readers familiar with optimization, the weak point of the ℓ_0 minimization procedure is that the function one seeks to minimize — the **objective function** — is not convex.

Minimizing a nonconvex function, even over an affine space such as $x + \mathrm{Ker}(A)$, is a difficult problem in general. An idea that then comes naturally to mind (for those who are used to minimizing convex functions) is to convexify the objective function, which here is the ℓ_0 function $x \in \mathbb{R}^N \to \|x\|_0$. One is therefore led to look for the convex function closest to the ℓ_0 function. A function of this type is called a *convex envelope*, whose definition we now recall.

Definition 4.1. *([4]) Let $N \geq 1$ be an integer and let C be a convex subset of $\mathbb{R}^N$. Let $f : C \to \mathbb{R}$ be a function. The **convex envelope** of f is the largest convex function g such that $g(x) \leq f(x)$ for every $x \in C$. We denote by $\mathrm{conv}(f)$ the convex envelope of f.*

This definition is meaningful: if $\mathcal{C}$ denotes the set of all convex functions lying below f, then $\sup_{g \in \mathcal{C}} g$ is itself a convex function, and is therefore equal to $\mathrm{conv}(f)$.

For the following result, we recall the notation B_p^N, which denotes the unit ball of the ℓ_p^N norm, that is, the set of all vectors $x \in \mathbb{R}^N$ such that $\|x\|_p \leq 1$.

Theorem 4.2. *The ℓ_1 norm is the convex envelope of the ℓ_0 function on B_∞^N, the unit ball of $(\mathbb{R}^N, \ell_\infty^N)$.*

Proof. We prove the result only on B_1^N (the unit ball of $(\mathbb{R}^N, \ell_1^N)$). To obtain the result on B_∞^N, one appeals to the convex biconjugate of ℓ_0, a notion that will be introduced in the chapter on matrix completion (the interested reader is referred to [4]). For the moment, we give a simple proof on B_1^N.

Let F be the convex envelope of $\ell_0 : x \to \|x\|_0$ on B_1^N. We show that F is the ℓ_1^N norm.

1) For every $x \in B_1^N$, one has $\|x\|_1 \leq \|x\|_0$, and the map $x \mapsto \|x\|_1$ is convex. Hence, by definition of the convex envelope of ℓ_0 on B_1^N, one has $\|x\|_1 \leq F(x)$ for every $x \in B_1^N$. We now prove the reverse inequality.

2) Since $\|x\|_1 \leq F(x)$ for every $x \in B_1^N$, one has in particular, writing $(e_1, \ldots, e_N)$ for the canonical basis of $\mathbb{R}^N$, that for every $i = 1, \ldots, N$, $1 = \|\pm e_i\|_1 \leq F(\pm e_i) \leq \|\pm e_i\|_0 = 1$, and therefore $F(\pm e_i) = 1$. Likewise, $F(0) = 0$, because $0 = \|0\|_1 \leq F(0) \leq \|0\|_0 = 0$. Let $x = (x_1, \ldots, x_N)^\top \in \mathbb{R}^N$ be such that $x \in B_1^N$, that is, such that $\sum_{j=1}^N |x_i| \leq 1$. By convexity of F and the relation $F(0) = 0$, one has

$$\begin{aligned} \|x\|_1 \leq F(x) &= F\Big(\sum_{j=1}^N |x_j| \mathrm{sign}(x_j) e_j + (1 - \|x\|_1) 0 \Big) \\ &\leq \sum_{j=1}^N |x_j| F(\mathrm{sign}(x_j) e_j) + (1 - \|x\|_1) F(0) = \sum_{j=1}^N |x_j| = \|x\|_1 . \end{aligned} \tag{4.1}$$

Hence, for every $x \in B_1^N$, $F(x) = \|x\|_1$. This proves that the convex envelope of ℓ_0 on B_1^N is indeed the ℓ_1^N norm. ∎

Remark 4.3. *The convex envelope of ℓ_0 over the whole of $\mathbb{R}^N$ is the function identically equal to zero, which carries very little information. This is why, here, we consider the convex envelope of ℓ_0 only on B_∞^N or on B_1^N. The choice is nevertheless arbitrary, but it allows us to view the ℓ_1^N norm as a convex relaxation of ℓ_0. There are other ways of introducing the ℓ_1 norm for this problem.*

It is therefore through convex relaxation that the ℓ_1 norm enters the Compressed Sensing problem. More generally, the ℓ_1 norm plays a central role in high-dimensional statistics. Here, for the CS problem, the procedure that has been studied the most is known under the name of Basis Pursuit.

Definition 4.4 ([3, 1])**.** *Given a matrix $A \in \mathbb{R}^{m \times N}$ and a vector $y \in \mathbb{R}^m$, the **Basis Pursuit** (BP) procedure returns*

$$\hat{x} \in \operatorname*{argmin}_{t \in \mathbb{R}^N : At = y} \|t\|_1 , \tag{BP}$$

if there exists a solution of the system $Ax = y$, and $\emptyset$ otherwise.

Problem (BP) is a convex relaxation of the ℓ_0 minimization problem, obtained by convexifying the objective function (here the ℓ_0 "norm"). Before turning to the theoretical study of this procedure (that is, to the determination of conditions on A under which the procedure reconstructs exactly every vector in Σ_s, and to the determination of the minimal number of measurements together with matrices A satisfying these conditions), we first consider its algorithmic properties, which were the principal defect of ℓ_0 minimization.

Conclusion: Convex relaxation is a tool widely used in high-dimensional statistics. In some cases, it turns problems that admit no reasonable algorithmic solution into procedures that can be implemented efficiently, thanks to the tools of convex optimization. It is therefore an essential tool to master, and a useful reflex when one seeks to recover structures which seem, for instance, to require a combinatorial search through sets of large cardinality.

Remark 4.5. *Methods that use the ℓ_1 norm, either as an objective function to be minimized or as a regularization function, abound in high-dimensional statistics. They have been used empirically in a variety of fields, such as geology and geophysics ([2], Taylor et al. (1979), Levy and Fullager (1981), Oldenburg et al. (1983), Santosa and Symes (1988)), in radio astronomy (Högbom (1974), Schwarz (1978)), in spectroscopy (Kawata et al. (1983), Mammone (1983), Minami et al. (1985), Barkhuijsen (1985), Newman (1988)), in medical imaging (Papoulis and Chamzas (1979)), and elsewhere. Other celebrated methods based on the ℓ_1 norm are the LASSO, the Dantzig selector, and Matching Pursuit.*

4.1 A linear programming procedure for Basis Pursuit

The Basis Pursuit procedure is very popular in CS because, in particular, it has good algorithmic properties. Indeed, problem (BP) can be rewritten as a linear programming problem. Problems of this type are, in a sense, the easiest to solve in convex optimization: one minimizes a linear objective function subject to linear constraints.

Consider the following linear programming problem: given $A \in \mathbb{R}^{m\times N}$ and $y \in \mathbb{R}^m$,

$$\left(\widehat{z^+}, \widehat{z^-}\right) \in \operatorname*{argmin}_{(z^+,z^-)\in\mathbb{R}^{2N}} \sum_{j=1}^{N} z_j^+ + z_j^- \text{ subject to } [A|-A]\begin{bmatrix} z^+ \\ z^- \end{bmatrix} = y \text{ and } \begin{bmatrix} z^+ \\ z^- \end{bmatrix} \geq 0. \tag{LP}$$

We recall that $u \geq 0$, for $u \in \mathbb{R}^{2N}$, means that every coordinate of u is nonnegative.

Problem (LP) is a convex optimization problem in $\mathbb{R}^{2N}$ whose objective function is linear, as are the constraints. It is therefore a linear programming problem. We show below that problems (BP) and (LP) are equivalent. We first recall the following notation: for every $x \in \mathbb{R}^N$, one denotes by $x^+ \in \mathbb{R}^N$ and $x^- \in \mathbb{R}^N$ the two vectors in $\mathbb{R}^N$ whose coordinates are given, for every $j = 1, \ldots, N$, by

$$(x^+)_j = \max(0, x_j) \text{ and } (x^-)_j = \max(0, -x_j).$$

In particular, one has $x = x^+ - x^-$.

Proposition 4.6. *The two problems (BP) and (LP) are equivalent:*

1. *If $\hat{x}$ is a solution of (BP), then $(\hat{x}^+, \hat{x}^-)$ is a solution of (LP).*
2. *If $\left(\widehat{z^+}, \widehat{z^-}\right)$ is a solution of (LP), then $\widehat{z^+} - \widehat{z^-}$ is a solution of (BP).*

Proof. Assume that $\hat{x}$ is a solution of (BP), and consider its positive part $\hat{x}^+$ and its negative part $\hat{x}^-$. One has

$$[A|-A]\begin{bmatrix} \hat{x}^+ \\ \hat{x}^- \end{bmatrix} = A(\hat{x}^+ - \hat{x}^-) = A\hat{x} = y \text{ and } \begin{bmatrix} \hat{x}^+ \\ \hat{x}^- \end{bmatrix} \geq 0 \tag{4.2}$$

so that $(\hat{x}^+, \hat{x}^-)$ does lie in the constraint set of (LP). It remains only to show that $\sum_{j=1}^N \hat{x}_j^+ + \hat{x}_j^-$ is minimal over this set.

To this end, take $(z^+, z^-) \in \mathbb{R}^{2N}$ in the constraint set of (LP). Write $z = z^+ - z^-$. Then $y = [A|-A]\begin{bmatrix} z^+ \\ z^- \end{bmatrix} = A(z^+ - z^-) = Az$, so z lies in the constraint set of (BP). By minimality of the ℓ_1 norm of $\hat{x}$ over the constraint set of (BP) (to which z belongs), one has

$$\sum_{j=1}^{N} \hat{x}_j^+ + \hat{x}_j^- = \|\hat{x}\|_1 \leq \|z\|_1 = \sum_{j=1}^{N} |z_j^+ - z_j^-| \leq \sum_{j=1}^{N} z_j^+ + z_j^-$$

since $z_j^+, z_j^- \geq 0$ for every $j = 1, \ldots, N$. As this holds for every z in the constraint set of (LP), we conclude that $(\hat{x}^+, \hat{x}^-)$ is a solution of (LP).

Conversely, let $\left(\widehat{z^+}, \widehat{z^-}\right) \in \mathbb{R}^{2N}$ be a solution of (LP). Write $z = \widehat{z^+} - \widehat{z^-}$. Since $y = [A|-A]\begin{bmatrix} \widehat{z^+} \\ \widehat{z^-} \end{bmatrix} = A\left(\widehat{z^+} - \widehat{z^-}\right) = Az$, the vector z lies in the constraint set of (BP). Moreover, by the same argument as in (4.2), if $x \in \mathbb{R}^N$ lies in the constraint set of (BP), then (x^+, x^-) lies in that of (LP), and therefore

$$\|z\|_1 = \sum_{j=1}^{N} \left|\widehat{z^+}_j - \widehat{z^-}_j\right| \leq \sum_{j=1}^{N} \widehat{z^+}_j + \widehat{z^-}_j \leq \sum_{j=1}^{N} x_j^+ + x_j^- = \|x\|_1 .$$

Thus z has minimal ℓ_1 norm over the constraint set of (BP), and is a solution of that problem. ■

Proposition 4.6 therefore guarantees that the Basis Pursuit procedure can be used in practice. The situation is in fact even more favorable, since (BP) can be rewritten as a linear program, a convenient framework in convex optimization. Convex relaxation has thus taken us from an NP-hard problem, (P_0), to a linear programming problem, (LP). This is an important gain from the computational point of view. One might now think that so large a gain on the practical side must be paid for on the theoretical side. We shall show that this is not really the case. We therefore turn to the theoretical properties of (BP).

4.2 Minimal number of measurements for Basis Pursuit

It is hard to imagine that the very substantial computational gain achieved by convex relaxation — the "transformation" of an NP-hard problem into a linear programming problem — is not "paid for" at some other level. In what follows, we therefore study the properties of Basis Pursuit and, in particular, the property of greatest interest to us: that of solving the compressed sensing problem on Σ_s.

We saw in the previous sections that the ℓ_0 minimization procedure can solve the CS problem with only $m = 2s$ measurements (under certain conditions on the measurement matrix). We would now like to know whether Basis Pursuit can also solve this problem, and to determine the minimal number of measurements that are necessary and sufficient for it to do so. We formalize this property in the following definition.

Definition 4.7. *Let $A : \mathbb{R}^N \to \mathbb{R}^m$ be a matrix with $m \leq N$, and let s be an integer smaller than N. One says that A satisfies the property of* ***exact reconstruction of order*** **s** *when, for every s-sparse vector x,*

$$\operatorname*{argmin}_{t \in \mathbb{R}^N} \left(\|t\|_1 : At = Ax \right) = \{x\}.$$

One then says that A satisfies $RE(s)$.

The property of exact reconstruction of order s therefore characterizes those measurement matrices A that reconstruct exactly every s-sparse vector x from the datum Ax, by means of the Basis Pursuit algorithm. One of the aims of this course is to determine the matrices satisfying the exact reconstruction property of order s for a minimal number of measurements (that is, of rows). Later in the course, we shall see sufficient conditions for exact reconstruction of order s.

For the moment, one can already establish a lower bound on the number m of rows of any matrix satisfying this property. In other words, the exact reconstruction property of Definition 4.7 necessarily forces a minimal number of measurements. This result may be set alongside Proposition 2.2, which shows that the reconstruction property of the ℓ_0 minimization procedure on Σ_s implies $m \geq 2s$. Here, for the procedure (BP), slightly more measurements will be needed.

Proposition 4.8. *Let $A : \mathbb{R}^N \longmapsto \mathbb{R}^m$ be a matrix satisfying $RE(2s)$. Then*

$$m \geq \frac{1}{\log 5} \left\lfloor \frac{s}{2} \right\rfloor \log \left(\left\lfloor \frac{N}{8es} \right\rfloor \right).$$

Proof. Let $A : \mathbb{R}^N \longmapsto \mathbb{R}^m$. Consider the quotient vector space $\ell_1^N/\mathrm{Ker}(A)$, the quotient map, and the quotient norm:

$$Q : \ell_1^N \longmapsto \ell_1^N/\mathrm{Ker}(A) \text{ and } \|Qx\|_1 = \min_{h \in \mathrm{Ker}(A)} \|x + h\|_1 . \tag{4.3}$$

Recall that $\mathbb{R}^N/\mathrm{Ker}(A)$ is the set of all equivalence classes $Qx = x + \mathrm{Ker}(A)$, for $x \in \mathbb{R}^N$. We write $\ell_1^N/\mathrm{Ker}(A)$ for the space $\mathbb{R}^N/\mathrm{Ker}(A)$ equipped with the quotient norm introduced in (4.3). Intuitively, one may think of $\ell_1^N/\mathrm{Ker}(A)$ as the vector space $\mathrm{Im}(A^\perp)$ equipped with a certain norm. Indeed, since $\mathrm{Im}(A^\top) = (\mathrm{Ker}(A))^\perp$ (because if $z \in \mathrm{Im}(A^\top)$, then for every $x \in \mathrm{Ker}(A)$ one has $\langle x, z \rangle = \langle x, A^\perp y \rangle = \langle Ax, y \rangle = 0$, hence $\mathrm{Im}(A^\top) \subset (\mathrm{Ker}(A))^\perp$, and by the rank theorem $\dim(\mathrm{Im}(A^\top)) = \mathrm{rg}(A^\top) = \mathrm{rg}(A) = N - \dim(\mathrm{Ker}(A)) = \dim((\mathrm{Ker}(A))^\perp)$), one has $\mathbb{R}^N = \mathrm{Ker}(A) \oplus^\perp \mathrm{Im}(A^\top)$, and every equivalence class Qx may then be represented by a unique element of $\mathrm{Im}(A^\top)$, namely the component of x in $\mathrm{Im}(A^\top)$. In particular, $\dim\left(\ell_1^N/\mathrm{Ker}(A)\right) = \dim(\mathrm{Im}(A^\top)) = \mathrm{rg}(A^\top) \leq m$.

If A satisfies $RE(2s)$, then for every $x \in \Sigma_{2s}$ one has $\|Qx\|_1 = \|x\|_1$, that is, Q preserves norms on Σ_{2s}. Since $\Sigma_s - \Sigma_s \subset \Sigma_{2s}$, one has, for all $x, y \in \Sigma_s$,

$$\|Qx - Qy\|_1 = \|x - y\|_1 . \tag{4.4}$$

That is, Q is an isometry on Σ_s. This isometry will be used to transport the metric complexity of the set (Σ_s, ℓ_1^N) into the quotient space $\ell_1^N/\mathrm{Ker}(A)$, whose dimension is at most m (much smaller than the ambient dimension N), and thereby to deduce a lower bound on m. Indeed, in view of the isometry property of Q on Σ_s, the unit ball of $\ell_1^N/\mathrm{Ker}(A)$ must contain a 1/2-separated set of large cardinality.

We begin by studying the complexity of (Σ_s, ℓ_1^N). We use the following lemma, which can be obtained by a successive enumeration, for instance, and whose proof is given in the accompanying corrected exercise sheet.

Lemma 4.9 (Varshamov–Gilbert)**.** *Let $s \leq N/2$. There exists a family $\mathcal{S}$ of subsets of $\{1, \ldots, N\}$ such that*

1. *$\forall S \in \mathcal{S}$, $|S| = s$,*
2. *$\forall S_1, S_2 \in \mathcal{S}$, $S_1 \neq S_2 \Rightarrow |S_1 \cap S_2| \leq \lfloor s/2 \rfloor$,*

3. $\log|\mathcal{S}| \geq \lfloor \frac{s}{2} \rfloor \log\left(\lfloor \frac{N}{8es} \rfloor\right)$.

Proof of Proposition 4.8. Consider a family $\mathcal{S}$ as in Lemma 4.9 and, for every $S \in \mathcal{S}$, write $x(S) = s^{-1}\sum_{i\in S} e_i \in B_1^N$. For all $S_1 \neq S_2 \in \Sigma_s$, one has

$$\|x(S_1) - x(S_2)\|_1 = \left\| \frac{1}{s} \sum_{i \in S_1 \Delta S_2} e_i \right\|_1 = \frac{|S_1 \Delta S_2|}{s} \geq 1 > \frac{1}{2}$$

where $S_1 \Delta S_2 = S_1 \cup S_2 - S_1 \cap S_2$.

Moreover, for every $S \in \mathcal{S}$, $\|Qx(S)\|_1 = \|x(S)\|_1 = 1$, so that $\{x(S) : S \in \mathcal{S}\}$ is a subset of the unit ball of $\ell_1^N/\mathrm{Ker}(A)$. It follows from the isometry property (4.4) that the unit ball of $\ell_1^N/\mathrm{Ker}(A)$ contains a set that is 1/2-separated for its natural norm. This set has cardinality $|\mathcal{S}|$. The cardinality of any such set may be bounded from above by a volumetric argument, as follows.

Lemma 4.10 (Volumetric argument)**.** *Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Write B for its unit ball. Let $0 < \epsilon \leq 1$ and let $\Lambda \subset B$ be such that $\|x - y\| > \epsilon$ for all $x, y \in \Lambda$. Then the cardinality of Λ necessarily satisfies*

$$|\Lambda| \leq \left(1 + \frac{2}{\epsilon}\right)^n$$

Thus, by the volumetric argument of Lemma 4.10, any such set has cardinality at most $5^{\mathrm{rank}A}$ (where $\mathrm{rank}A = \dim\big(\ell_1^N/\mathrm{Ker}(A)\big)$). Hence $|\mathcal{S}| \leq 5^{\mathrm{rank}A} \leq 5^m$, and the result follows from the lower bound on the cardinality $|\mathcal{S}|$ in Lemma 4.9. ∎

Remark 4.11. *When $N \geq (16e)^2 es$ and $s \geq 4$, one has*

$$\frac{1}{\log 5} \left\lfloor \frac{s}{2} \right\rfloor \log\left(\left\lfloor \frac{N}{8es} \right\rfloor \right) \geq \frac{s}{8 \log 5} \log\left(\frac{eN}{s}\right).$$

In the sequel, we shall therefore say that the minimal number of measurements needed for a matrix to satisfy ER(s) is of order $s \log(eN/s)$.

Conclusion: In order to reconstruct every s-sparse vector x from m measurements Ax by means of Basis Pursuit, one must necessarily have m larger than $s\log(eN/s)$, up to an absolute constant. We shall see later that there do exist matrices satisfying $RE(2s)$ with m of order $s\log(eN/s)$. In particular, one "loses" only a logarithmic factor by comparison with the condition "$m \geq 2s$", which is necessary (and sufficient for certain matrices) in order that the ℓ_0 minimization procedure solve the CS problem on Σ_s. The very substantial computational gain obtained by convex relaxation is therefore paid for only by a logarithmic loss in the number of measurements, which is almost nothing.

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